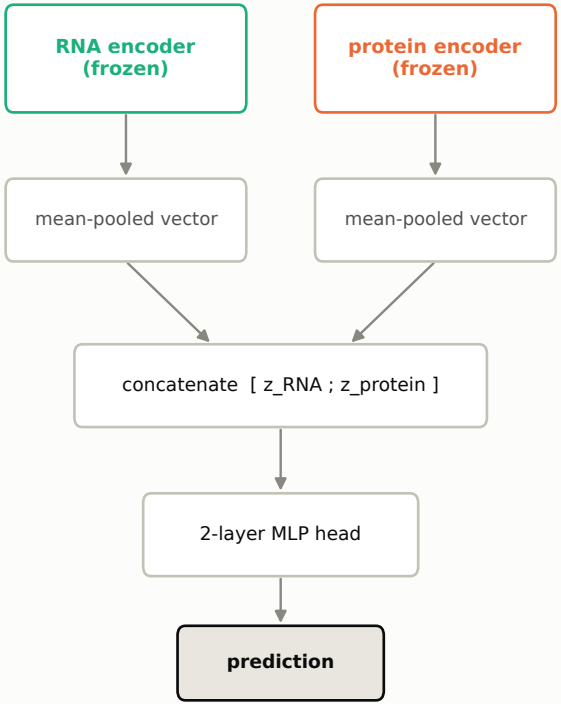
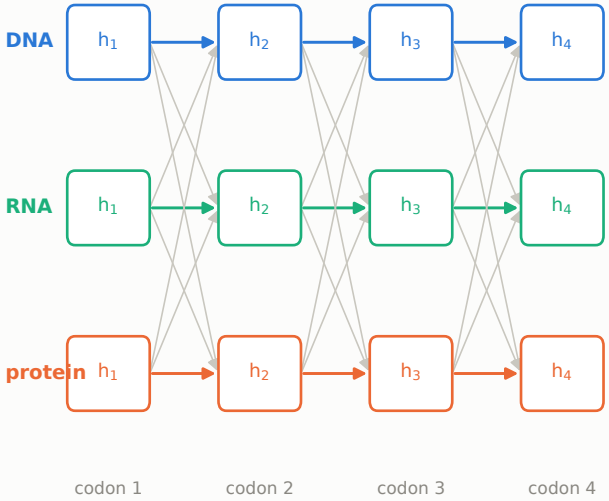
a Concatenation + MLP (reference baseline B4)

no token interaction; sets the bar every fusion model must clear



b Coupled Mamba on the codon grid (design 1)

one recurrent chain per modality, coupled at every codon step t

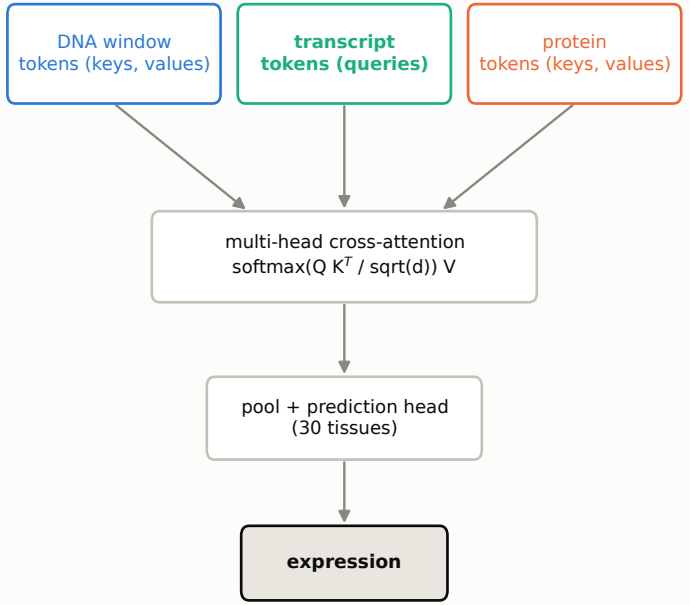


$h_t^m = S_m$ (sum over modalities of h_{t-1}) + $B_m\, x_t^m$

linear in sequence length; no alignment loss; needs the shared codon grid

c Cross-attention, IsoFormer style (design 2)

token streams keep their own lengths; one stream queries the others



quadratic in length; no shared grid needed; also the benchmark competitor on stability